Binomial Distribution

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Conditions for a binomial distribution**

State the four conditions required for a random variable to follow a binomial distribution.

(1) A fixed number of trials n. (2) Each trial has only two outcomes (success or failure). (3) The probability of success p is constant for every trial. (4) The trials are independent of each other.

2 **[Foundational] • Target: Conditions for a binomial distribution**

A survey asks 50 people whether they support a new policy, and X is the number who say yes, assuming each person's response is independent with the same probability of support. Explain why X follows a binomial distribution.

X follows a binomial distribution because there is a fixed number of trials ($n = 50$), each trial (person) has two outcomes (yes or no), the probability of a 'yes' response is constant across all people, and the responses are independent of each other — satisfying all four binomial conditions.

3 **[Foundational] • Target: Conditions for a binomial distribution**

A researcher rolls a die 15 times and counts the number of times a 3 appears. Explain why this count follows a binomial distribution.

This count follows a binomial distribution because there is a fixed number of trials ($n = 15$), each roll has two outcomes (rolling a 3 or not), the probability of rolling a 3 is constant ($\frac{1}{6}$) on every roll, and the rolls are independent of each other — satisfying all four binomial conditions.

4 **[Foundational] • Target: Calculating binomial probabilities**

A fair coin is tossed 8 times. Find the probability of getting exactly 5 heads, using $P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$.

$X \sim B(8, 0.5)$. $P(X = 5) = \binom{8}{5}(0.5)^5(0.5)^3 = 56(0.03125)(0.125) = 56(0.00390625) \approx 0.2188$.

5 **[Foundational] • Target: Calculating binomial probabilities**

A factory produces items with a defect rate of 10%. In a sample of 6 items, find the probability that exactly 2 are defective.

$X \sim B(6, 0.1)$. $P(X = 2) = \binom{6}{2}(0.1)^2(0.9)^4 = 15(0.01)(0.6561) \approx 0.0984$.

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6 [Foundational] • Target: Calculating binomial probabilities

A fair coin is tossed 6 times. Find the probability of getting exactly 4 heads.

$X \sim B(6, 0.5)$. $P(X = 4) = \binom{6}{4}(0.5)^4(0.5)^2 = 15(0.0625)(0.25) = 0.234375$.

7 [Foundational] • Target: Calculating binomial probabilities

A factory produces items with a defect rate of 20%. In a sample of 5 items, find the probability that exactly 1 is defective.

$X \sim B(5, 0.2)$. $P(X = 1) = \binom{5}{1}(0.2)^1(0.8)^4 = 5(0.2)(0.4096) = 0.4096$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Cumulative binomial probabilities

A biased coin has $P(\text{heads}) = 0.3$. It is tossed 5 times. Find $P(X \leq 1)$, where X is the number of heads.

$P(X = 0) = \binom{5}{0}(0.3)^0(0.7)^5 = 0.16807$. $P(X = 1) = \binom{5}{1}(0.3)^1(0.7)^4 = 5(0.3)(0.2401) = 0.36015$.
$P(X \leq 1) = 0.16807 + 0.36015 \approx 0.5282$.

9 [Intermediate] • Target: Cumulative binomial probabilities

Using the same distribution $X \sim B(5, 0.3)$, find $P(X \geq 2)$.

$P(X \geq 2) = 1 - P(X \leq 1) = 1 - 0.5282 \approx 0.4718$.

10 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(40, 0.25)$. Find the mean and variance, using $E(X) = np$ and $\text{Var}(X) = np(1 - p)$.

$E(X) = np = 40(0.25) = 10$. $\text{Var}(X) = np(1 - p) = 40(0.25)(0.75) = 7.5$.

11 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(60, 0.2)$. Find the mean and variance.

$E(X) = np = 60(0.2) = 12$. $\text{Var}(X) = np(1 - p) = 60(0.2)(0.8) = 9.6$.

12 [Intermediate] • Target: Finding an unknown parameter

A random variable $X \sim B(n, 0.4)$ has mean $E(X) = 20$. Find n.

$E(X) = np$: $20 = n(0.4)$, so $n = 50$.

13 [Intermediate] • Target: Finding an unknown parameter

A random variable $X \sim B(n, 0.25)$ has variance $\text{Var}(X) = 9$. Find n.

$\text{Var}(X) = np(1 - p)$: $9 = n(0.25)(0.75) = 0.1875n$, so $n = \frac{9}{0.1875} = 48$.

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14 **[Intermediate]** • **Target: Using the binomial distribution to model a real scenario**

A multiple-choice quiz has 10 questions, each with 4 options, only one of which is correct. A student guesses randomly on every question. Find the probability the student gets at least 8 questions correct.

$X \sim B(10, 0.25)$. $P(X = 8) = \binom{10}{8}(0.25)^8(0.75)^2 \approx 45(0.0000153)(0.5625) \approx 0.000387$.

$P(X = 9) = \binom{10}{9}(0.25)^9(0.75)^1 \approx 10(0.00000381)(0.75) \approx 0.0000286$. $P(X = 10) = (0.25)^{10} \approx 0.00000095$.

$P(X \geq 8) \approx 0.000387 + 0.0000286 + 0.00000095 \approx 0.000417$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Using the binomial distribution to model a real scenario** • **Cross-Lesson: 14.01 Energy, Work & Power**

A basketball player has a 70% free-throw success rate. Find the probability that the player makes exactly 3 out of 5 free throws.

$X \sim B(5, 0.7)$. $P(X = 3) = \binom{5}{3}(0.7)^3(0.3)^2 = 10(0.343)(0.09) = 0.3087$.

16 **[Challenging]** • **Target: Using the binomial distribution to model a real scenario** • **Cross-Lesson: 15.01 Representation of Data**

An archer hits the target with probability 0.8 on each shot, independently. Find the probability that the archer hits the target exactly 4 times out of 5 shots.

$X \sim B(5, 0.8)$. $P(X = 4) = \binom{5}{4}(0.8)^4(0.2)^1 = 5(0.4096)(0.2) = 0.4096$.

17 **[Challenging]** • **Target: Distinguishing binomial situations from non-binomial situations** • **Cross-Lesson: 16.01 Permutations & Combinations**

Explain why drawing 5 cards from a standard deck WITHOUT replacement and counting the number of aces drawn is NOT a binomial distribution, while drawing 5 cards WITH replacement would be.

In a binomial distribution, the probability of success must remain constant across all trials, which requires independence between trials. When drawing without replacement, the probability of drawing an ace changes after each draw (since the deck composition changes), so the trials are not independent and p is not constant. Drawing with replacement restores this constant probability and independence, satisfying the binomial conditions.

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18 **[Challenging] • Target: Distinguishing binomial situations from non-binomial situations • Cross-Lesson: 17.01 Discrete Random Variables**

Explain why the number of sixes rolled in 10 rolls of a fair die IS a binomial distribution, while the number of rolls needed to get the first six is NOT.

The number of sixes in 10 rolls is binomial because there is a fixed number of trials (10), each roll has two outcomes (six or not six), the probability of a six is constant ($\frac{1}{6}$) each time, and the rolls are independent. The number of rolls needed to get the first six does not have a fixed number of trials — the trial count itself is the random variable — so it does not satisfy the binomial conditions (it follows a geometric distribution instead).

19 **[Challenging] • Target: Distinguishing binomial situations from non-binomial situations • Cross-Lesson: 19.01 Normal Distribution**

Explain why the number of heads obtained by flipping a fair coin exactly 20 times IS a binomial distribution, while the number of flips needed to get 3 heads is NOT.

The number of heads in 20 flips is binomial because there is a fixed number of trials (20), each flip has two outcomes (heads or tails), the probability of heads is constant (0.5) on every flip, and the flips are independent. The number of flips needed to get 3 heads does not have a fixed number of trials — the trial count itself is variable — so it does not satisfy the binomial conditions (it follows a negative binomial distribution instead).

20 **[Challenging] • Target: A well-designed word problem: quality control sampling • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities**

This question has two parts. A machine produces bolts, of which 5% are defective. A quality inspector randomly samples 20 bolts each hour. (a) Find the expected number of defective bolts in a sample of 20. (b) The inspector rejects the whole batch if 3 or more defective bolts are found in the sample. Find the probability that the batch is rejected, given $P(X = 0) \approx 0.3585$, $P(X = 1) \approx 0.3774$, and $P(X = 2) \approx 0.1887$ for $X \sim B(20, 0.05)$.

(a) $E(X) = np = 20(0.05) = 1$ defective bolt. (b) $P(X \geq 3) = 1 - P(X \leq 2) = 1 - [0.3585 + 0.3774 + 0.1887] = 1 - 0.9246 = 0.0754$, so there is approximately a 7.5% chance the batch is rejected.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Conditions for a binomial distribution

A machine produces metal rods whose lengths are normally distributed with unknown mean μ cm and known standard deviation $\sigma = 0.8$ cm. A quality control manager tests the hypotheses $H_0 : \mu = 25.0 \quad \text{against} \quad H_1 : \mu < 25.0$ at the 5% level of significance. A random sample of 16 rods yields a sample mean length of $\bar{x} = 24.6$ cm. Given that the critical value for this test is $z_{\text{crit}} = -1.645$, which of the following gives the correct test statistic z and the corresponding conclusion?

a. $z = -2.0$**; reject** H_0 **because** $-2.0 < -1.645$ ✓

b. $z = -2.0$; do not reject H_0 because $-2.0 < -1.645$

c. $z = -0.5$; do not reject H_0 because $-0.5 > -1.645$

d. $z = -8.0$; reject H_0 because $-8.0 < -1.645$

1. Under the null hypothesis $H_0 : \mu = 25.0$, the sample mean $\bar{X}$ of a random sample of size $n = 16$ from $\text{N}(\mu, \sigma^2)$ has standard error: $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{16}} = \frac{0.8}{4} = 0.2$ 2. The standardized test statistic z is: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{24.6 - 25.0}{0.2} = \frac{-0.4}{0.2} = -2.0$ 3. Since this is a lower-tail test at the 5% level with critical value $z_{\text{crit}} = -1.645$, the critical (rejec.

22 [Foundational] • Target: Calculating binomial probabilities

The unique real root α of the equation $x^3 + 2x - k = 0,$ where k is a positive constant, is estimated using the iterative formula $x_{n+1} = \frac{k}{x_n^2 + 2}$. The sequence of iterations converges locally to α (for an initial approximation x_0 sufficiently close to α) if and only if $|g'(\alpha)| < 1$, where $g(x) = \frac{k}{x^2 + 2}$. Determine the complete set of values of k for which this iteration converges locally to α.

a. $0 < k < 4\sqrt{2}$ ✓

b. $0 < k < 2\sqrt{2}$

c. $0 < k < 3$

d. $0 < k < 12$

To determine the range of k for which the iteration converges locally, we follow these steps: 1. **Root relation:** At the fixed point $x = \alpha$, we have: $\alpha = g(\alpha) = \frac{k}{\alpha^2 + 2} \Rightarrow \alpha^3 + 2\alpha = k.$ Since $k > 0$ and the function $f(x) = x^3 + 2x - k$ satisfies $f'(x) = 3x^2 + 2 > 0$ for all real x, there is a unique real root α, and $\alpha > 0$. 2. **Differentiating $g(x)$:** Using the chain rule on $g(x) = k(x^2 + 2)^{-1}$.

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23 [Foundational] • Target: Cumulative binomial probabilities

A random sample of $n = 50$ observations is taken from a population. The sample values are summarised as follows: $\sum x = 450, \quad \sum x^2 = 4295$. Find an unbiased estimate of the population variance.

a. **5.0** ✓

b. 4.9

c. 2.24

d. 9.0

The unbiased estimate of the population variance, denoted s^2, is calculated using the formula: $s^2 = \frac{1}{n-1}\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)$ Substitute the given values $n = 50$, $\sum x = 450$, and $\sum x^2 = 4295$: $s^2 = \frac{1}{50-1}\left(4295 - \frac{450^2}{50}\right)$ $s^2 = \frac{1}{49}\left(4295 - \frac{202500}{50}\right)$ $s^2 = \frac{1}{49}(4295 - 4050) = \frac{245}{49} = 5.0$ Why other choices fail: - 4.9 is the biased sample variance obtained by dividing by.

24 [Foundational] • Target: Mean and variance of a binomial distribution

A machine produces steel rods whose lengths are normally distributed with standard deviation $\sigma = 2.4$ cm. A quality controller suspects that the mean length has increased from the target value of 50 cm. A random sample of 36 rods is measured, yielding a sample mean of $\bar{x} = 50.8$ cm. Find the value of the standardized test statistic z for testing $H_0 : \mu = 50$ against $H_1 : \mu > 50$.

a. 2.0 ✓

b. 0.33

c. 1.2

d. 0.5

Under the null hypothesis $H_0 : \mu = 50$, the sample mean of $n = 36$ independent observations follows a normal distribution: $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N\left(50, \frac{2.4^2}{36}\right)$ The standard error of the mean is: $\text{SE} = \frac{\sigma}{\sqrt{n}} = \frac{2.4}{\sqrt{36}} = \frac{2.4}{6} = 0.4$ The standardized test statistic z is calculated as: $z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{50.8 - 50}{0.4} = \frac{0.8}{0.4} = 2.0$ Common errors include: - O.

25 [Foundational] • Target: Finding an unknown parameter

Formulate the null and alternative hypotheses for the following scenario: "A factory claims the mean weight of a product is 250g. An inspector wants to test if it has changed (either direction)."

a. $H_0 : \mu = 500,\ H_1 : \mu < 500$

b. $H_0 : \mu = 3,\ H_1 : \mu > 3$

c. $H_0 : \mu = 10,\ H_1 : \mu < 10$

d. $H_0 : \mu = 250,\ H_1 : \mu e 250$ ✓

Applying Hypothesis Testing principles, $H_0 : \mu = 250,\ H_1 : \mu e 250$. Therefore, the verified correct option is $H_0 : \mu = 250,\ H_1 : \mu e 250$.

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26 [Foundational] • Target: Using the binomial distribution to model a real scenario

Formulate the null and alternative hypotheses for the following scenario: "A manufacturer claims the mean lifetime of a battery is 500 hours. A researcher suspects it is less than this."

a. $H_0 : \mu = 500,\ H_1 : \mu < 500$ ✓

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 3,\ H_1 : \mu > 3$

d. $H_0 : \mu = 250,\ H_1 : \mu e 250$

Applying Hypothesis Testing principles, $H_0 : \mu = 500,\ H_1 : \mu < 500$. Therefore, the verified correct option is $H_0 : \mu = 500,\ H_1 : \mu < 500$.

27 [Foundational] • Target: Distinguishing binomial situations from non-binomial situations

Formulate the null and alternative hypotheses for the following scenario: "A school claims the mean test score is 70. A teacher believes it is actually higher."

a. $H_0 : \mu = 10,\ H_1 : \mu < 10$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$ ✓

c. $H_0 : \mu = 500,\ H_1 : \mu < 500$

d. $H_0 : \mu = 3,\ H_1 : \mu > 3$

Applying Hypothesis Testing principles, $H_0 : \mu = 70,\ H_1 : \mu > 70$. Therefore, the verified correct option is $H_0 : \mu = 70,\ H_1 : \mu > 70$.

28 [Foundational] • Target: A well-designed word problem: quality control sampling

A confidence interval for the mean uses $z = 1$, $\sigma = 4$, $n = 9$. Find the margin of error $E = z\sigma/\sqrt{n}$.

a. $\frac{7}{3}$

b. $\frac{4}{3}$ ✓

c. $\frac{17}{12}$

d. $\frac{8}{3}$

Applying Sampling & Estimation principles, $E = 1 \times 4/3 = \frac{4}{3}$. Therefore, the verified correct option is $\frac{4}{3}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Conditions for a binomial distribution

A sample of size $n = 9$ is drawn from a population with mean $\mu = 42$ and $\sigma = 8$. Given $\bar{x} = \frac{118}{3}$, find $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$.

a. 0

b. 1

c. **-1** ✓

d. -3

$\sigma/\sqrt{n} = 8/3 = \frac{8}{3}$. $\bar{x} - \mu = -\frac{8}{3}$. $Z = -\frac{8}{3} \div \frac{8}{3} = -1$.

30 [Intermediate] • Target: Calculating binomial probabilities

A sample of size 6 has $\sum x = 282$ and $\sum x^2 = 13312$. Find the unbiased estimate of the population variance.

a. $\frac{6656}{3}$

b. $\frac{58}{5}$ ✓

c. $\frac{63}{5}$

d. $\frac{29}{3}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{13312 - 282^2/6}{5} = \frac{58}{5}$.

31 [Intermediate] • Target: Cumulative binomial probabilities

$X \sim Po(3)$ and $Y \sim Po(3)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 9

b. 0

c. **6** ✓

d. 7

Since X and Y are independent Poisson variables, $X + Y \sim Po(3 + 3) = Po(6)$.

32 [Intermediate] • Target: Mean and variance of a binomial distribution

A sample of size 6 gives values: 5, 12, 18, 11, 17, 18. Find the sample mean $\bar{x}$.

a. 18

b. $\frac{29}{2}$

c. $\frac{27}{2}$ ✓

d. $\frac{81}{7}$

$\bar{x} = \frac{\sum x}{n} = \frac{81}{6} = \frac{27}{2}$.

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33 [Intermediate] • Target: Finding an unknown parameter

A confidence interval for the mean uses $z = 1$, $\sigma = 8$, $n = 9$. Find the total width of the confidence interval, $2E$.

a. $\frac{19}{3}$

b. $\frac{16}{3}$ ✓

c. 8

d. $\frac{8}{3}$

$E = \frac{8}{3}$, so width $= 2E = \frac{16}{3}$.

34 [Intermediate] • Target: Using the binomial distribution to model a real scenario

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.

a. **5.7** ✓

b. 6.7

c. 7.7

d. 4.7

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{5 \sin 80^\circ}{\sin 60^\circ} \approx 5.7$.

35 [Intermediate] • Target: Distinguishing binomial situations from non-binomial situations

The polynomial $P(x)$ is defined by $P(x) = (2 + ax)^n + bx^4 + cx - 4$ where n is an integer greater than 2, and a, b, c are constants with $a > 0$. In the binomial expansion of $(2 + ax)^n$ in ascending powers of x: - the coefficient of x^2 is twice the coefficient of x, - the coefficient of x^3 is equal to the coefficient of x^2. Given that $(x + 2)$ is a factor of $P(x)$, and that the remainder when $P(x)$ is divided by $(x + 1)$ is 7, find the remainder when $P(x)$ is divided by $(x - 1)$.

a. **1011** ✓

b. 1031

c. 1015

d. 499

We solve the problem in three stages. **Stage 1: Determine a and n using the Binomial Theorem** The general term in the expansion of $(2 + ax)^n$ is $\binom{n}{r} 2^{n-r} (ax)^r$. - The term in x is $\binom{n}{1} 2^{n-1} (ax) = n \cdot 2^{n-1} ax$. - The term in x^2 is $\binom{n}{2} 2^{n-2} (ax)^2 = \frac{n(n-1)}{2} 2^{n-2} a^2 x^2$. - The term in x^3 is $\binom{n}{3} 2^{n-3} (ax)^3 = \frac{n(n-1)(n-2)}{6} 2^{n-3} a^3 x^3$. Using the first condition (coefficient of x^2 is twice that of x).

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36 [Intermediate] • Target: A well-designed word problem: quality control sampling

Six-digit numbers are formed using digits chosen from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$, with no digit repeated in any number. Find the number of different six-digit odd numbers that can be formed such that the digits 2 and 4 are not adjacent.

a. 8160 ✓

b. 7680

c. 9120

d. 2880

We use complementary counting: $\text{Number of valid arrangements} = \text{Total odd 6 – digit numbers} - \text{Odd 6 – digit numbers where 2 and 4 are adjacent}$. **Step 1: Total number of 6-digit odd numbers** - For the number to be odd, the last digit (units place) must be chosen from the odd digits $\{1, 3, 5, 7\}$, giving 4 choices. - The remaining 5 positions are filled by selecting and arranging 5 digits from the remaining 7 digits in the set.

37 [Intermediate] • Target: Conditions for a binomial distribution

The real numbers x and y satisfy the simultaneous equations: $\log_3(x) + 2\log_9(y-1) = 2\ 4^y \cdot 8^{-\frac{x}{3}} = 32$ Find the value of $x^2 + y^2$.

a. 25 ✓

b. 10

c. $\frac{145}{4}$

d. 50

Step 1: Simplify the logarithmic equation using the change-of-base formula. Recall that $\log_9(y-1) = \frac{\log_3(y-1)}{\log_3(9)} = \frac{\log_3(y-1)}{2}$. Substituting this gives: $\log_3(x) + 2\left(\frac{\log_3(y-1)}{2}\right) = 2 \Rightarrow \log_3(x) + \log_3(y-1) = 2$ Applying the product rule of logarithms: $\log_3[x(y-1)] = 2 \Rightarrow x(y-1) = 3^2 = 9 \quad - - - \text{(Equation 1)}$ Step 2: Simplify the exponential equation using laws of indices. Express all terms wi.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Calculating binomial probabilities • Cross-Lesson: 17.01 Discrete Random Variables

In the expansion of $(1+ax)^n$ in ascending powers of x, the coefficient of x^2 is 60 and the coefficient of x^3 is 160, where n is a positive integer and $a>0$. A cubic polynomial $P(x)$, in which the coefficient of x^3 is 1, has $(x-a)^2$ as a factor and leaves a remainder of n when divided by $(x-1)$. Find the remainder when $P(x)$ is divided by $(x+a)$.

a. **48** ✓

b. -48

c. -112

d. 0

Step 1: Determine n and a using the Binomial Theorem. In the expansion of $(1+ax)^n$:

Term in $x^2 = \binom{n}{2}(ax)^2 = \frac{n(n-1)}{2}a^2x^2 = 60x^2 \Rightarrow n(n-1)a^2 = 120$ Term in $x^3 = \binom{n}{3}(ax)^3 = \frac{n(n-1)(n-2)}{6}a^3x^3 = 160x^3 \Rightarrow n(n-1)(n-2)a^3 = 960$ Dividing the equation for x^3 by the equation for x^2: $\frac{n(n-1)(n-2)a^3}{n(n-1)a^2} = \frac{960}{120} \Rightarrow (n-2)a = 8 \Rightarrow a = \frac{8}{n-2}$ Substi.

39 [Challenging] • Target: Cumulative binomial probabilities • Cross-Lesson: 19.01 Normal Distribution

Let the complex number z be represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Suppose z satisfies the equation $z^3 = -64i$. If z is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2} < \arg(z) < 2\pi$), find the real part of z.

a. 4

b. -4

c. $2\sqrt{3}$ ✓

d. $-2\sqrt{3}$

We are given the equation $z^3 = -64i$. First, express $-64i$ in polar form. The modulus is $|-64i| = 64$. The argument is $\arg(-64i) = \frac{3\pi}{2}$ (or $-\frac{\pi}{2}$). So, $-64i = 64\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)$. Let $z = r(\cos\phi + i\sin\phi)$. Then $z^3 = r^3(\cos(3\phi) + i\sin(3\phi))$. Equating the polar forms, we have $r^3 = 64$ and $3\phi = \frac{3\pi}{2} + 2k\pi$ for some integer k. From $r^3 = 64$, we get $r = 4$ (since r is a non-negative real number). Fro.

40 [Challenging] • Target: Mean and variance of a binomial distribution • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Determine the complete set of real values of the constant k for which the equation $2\cos 2\theta - 4\sin\theta = k$ has at least one real solution for θ.

a. $-6 \le k \le 3$ ✓

b. $-6 \le k \le 2$

c. $k \le 3$

d. $-3 \le k \le 6$

To find the set of values of k for which real solutions exist, we express the left-hand side entirely in terms of $\sin\theta$. Using the double-angle identity $\cos 2\theta = 1 - 2\sin^2\theta$: $2(1-2\sin^2\theta) - 4\sin\theta = k$ $-4\sin^2\theta - 4\sin\theta + 2 = k$ Let $u = \sin\theta$. Since $\theta \in \mathbb{R}$, u is constrained to the closed interval $[-1, 1]$. The problem reduces to finding the range of the quadratic function $f(u) = -4u^2 - 4u + 2, \quad u \in [-1, 1]$. 1. **Find the vertex (s.

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41 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 3.01 Graphs & Transformations

Find the set of all real solutions to the equation $\log_2(x-1) - \log_4(x+14) = 1$

a. $x = 11$ **only** ✓

b. $x = 11$ and $x = -5$

c. $x = 5$ only

d. $x = 16$ only

To solve the equation, we first express all logarithms in the same base using the change of base formula: $\log_4(x+14) = \frac{\log_2(x+14)}{\log_2 4} = \frac{\log_2(x+14)}{2} = \frac{1}{2}\log_2(x+14)$ Substitute this back into the original equation: $\log_2(x-1) - \frac{1}{2}\log_2(x+14) = 1$ Multiply through by 2: $2\log_2(x-1) - \log_2(x+14) = 2$ Apply the power law and the quotient law of logarithms.

42 [Challenging] • Target: Using the binomial distribution to model a real scenario • Cross-Lesson: 4.01 Coordinate Geometry

Determine the complete set of real values of the constant k for which the equation $e^x - kx^2 = 0$ has exactly three distinct real roots.

a. $k > \frac{e^2}{4}$ ✓

b. $k \geq \frac{e^2}{4}$

c. $0 < k < \frac{e^2}{4}$

d. $k > \frac{e}{2}$

First, note that $x = 0$ is not a solution to $e^x - kx^2 = 0$ since $e^0 = 1eq0$. Therefore, we can rewrite the equation as: $k = \frac{e^x}{x^2}$ Let $f(x) = \frac{e^x}{x^2}$ defined for all $xeq0$. The number of distinct real roots corresponds to the number of points of intersection between the curve $y = f(x)$ and the horizontal line $y = k$. Differentiating $f(x)$ using the quotient rule: $f'(x) = \frac{e^x \cdot x^2 - e^x \cdot 2x}{x^4} = \frac{e^x x(x-2)}{x^4} = \frac{e^x(x-2)}{x^3}$ Setting.

43 [Challenging] • Target: Distinguishing binomial situations from non-binomial situations • Cross-Lesson: 5.01 Trigonometry

The complex number z is defined by $z = \frac{a+3\mathrm{i}}{2+a\mathrm{i}}$, where a is a real constant. Given that $\arg(z) = \frac{\pi}{4}$, find the value of a.

a. 1 ✓

b. -6

c. 1 or -6

d. -1 or 6

To find a, we first express z in Cartesian form $x + \mathrm{i}y$ by multiplying the numerator and denominator by the complex conjugate of the denominator, $2 - a\mathrm{i}$: $z = \frac{(a+3\mathrm{i})(2-a\mathrm{i})}{(2+a\mathrm{i})(2-a\mathrm{i})} = \frac{2a - a^2\mathrm{i} + 6\mathrm{i} - 3a\mathrm{i}^2}{4+a^2} = \frac{5a + (6-a^2)\mathrm{i}}{4+a^2}$ Thus, the real and imaginary parts are: $\mathrm{Re}(z) = \frac{5a}{4+a^2}, \quad \mathrm{Im}(z) = \frac{6-a^2}{4+a^2}$ For.

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44 [Challenging] • Target: A well-designed word problem: quality control sampling • Cross-Lesson: 6.01 Differentiation & Integration

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers n. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.

a. 100 ✓

b. 110

c. 10

d. 210

Proof by Induction: Base Case ($n = 1$): $a_1 = 1$. The formula gives $1^2 = 1$. Thus, the base case holds. Inductive Hypothesis: Assume $a_k = k^2$ for some positive integer k. Inductive Step: We need to show that $a_{k+1} = (k+1)^2$. From the recurrence relation, $a_{k+1} = a_k + 2k + 1$. Using the inductive hypothesis, $a_{k+1} = k^2 + 2k + 1 = (k+1)^2$. Thus, by induction, $a_n = n^2$ for all positive integers n. Closed-form expression for S_N: $S_N = \sum_{n=1}^{N} a_n = \sum_{n=1}^{N} n^2$. This is the su.

45 [Challenging] • Target: Conditions for a binomial distribution • Cross-Lesson: 7.01 Statistics & Probability

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$ ✓

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

First, we factor the numerator and the denominator of $f(x) = \frac{x^3 - 8}{x^2 - 4}$. The numerator is a difference of cubes: $x^3 - 8 = (x-2)(x^2 + 2x + 4)$. The denominator is a difference of squares: $x^2 - 4 = (x-2)(x+2)$. Thus, $f(x) = \frac{(x-2)(x^2+2x+4)}{(x-2)(x+2)}$. A removable discontinuity occurs where a factor cancels out. Here, $(x-2)$ cancels, so the removable discontinuity is at $x = a = 2$. To find the limit as x approaches a, we consider the simplified function for $xeq2$.

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